

Session 2.2: Posterior Predictive Distribution and Monte Carlo computation

Imperial College London

Learning objectives

After this lecture you should be able to

- Describe what the posterior predictive distribution (PPD) is
- Explain how the PPD is computable
- Describe what Monte Carlo simulation is, and why it is useful
- The topic of posterior prediction is treated in Section 8.3 of Johnson, Ott, and Dogucu (2022)
- The topic of Monte Carlo simulation is presented in Sections 4.1-4.4 of Blangiardo and Cameletti (2015).

Outline

1. Bayesian predictive distribution
2. Computation of PPD
3. Introduction to Bayesian computing using Monte Carlo simulation
4. Example of MC computation

Posterior Predictive Distribution

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$$y \sim \text{Binomial}(\theta, n = 58)$$

- Let's θ be the true disease prevalence and y^* be the **predicted value**
- If θ were known, then we would predict

$$y^* | \theta \sim \text{Binomial}(30, \theta)$$

$$\text{thus } P(y \geq 5) = 1 - \left(\sum_{j=0}^4 \theta^j (1 - \theta)^{30-j} \right)$$

BUT ... θ is unknown

Source of variation in prediction:

- We don't know the true value of the parameters and we specify a prior on it:

$$\theta \sim \text{Beta}(a, b)$$

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To account for the sources of variation we iterate the following steps:

- 1 Sample from the posterior distribution $\theta \sim p(\theta \mid y)$
 - 2 Sample new values $y^* \sim p(y \mid \theta)$
- By repeating these steps a large number of times, we eventually obtain a reasonable approximation to the **posterior predictive distribution**.

Posterior Predictive Distribution (PPD)

- The **PPD** represents our uncertainty over the outcome of a future data collection, accounting for the observed data and model choice
- For the sake of prediction, the parameters are not of interest. They are vehicles by which the data inform about the predictive model
- The **PPD** averages over their posterior uncertainty

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta$$

- This properly accounts for parametric uncertainty
- The input is data, the output is a prediction distribution

Computation

Computing the PPD

- Say $\theta^{(1)}, \dots, \theta^{(M)}$ are samples from the posterior
- If we make a sample for y^* for each $\theta^{(m)}$,

$$y^{*(m)} \sim p(y|\theta^{(m)})$$

then the $y^{*(m)}$ are samples from the PPD

- The posterior predictive mean is approximated by the sample mean of the $y^{*(m)}$
- The probability that $y^* \geq 5$ is approximated by the sample proportion of the $y^{*(m)}$ that are equal or above 5

Example

Example

- We estimate the prevalence of a disease in the UK population using a sample of $n = 58$ individuals.
- We find that $y = 10$ individuals have the diseases.
- **What is the probability that, if we additionally sample ($k=30$) individuals this year, at least 5 will have the disease?**

1 Likelihood: $y \sim \text{Binomial}(\theta, 58)$

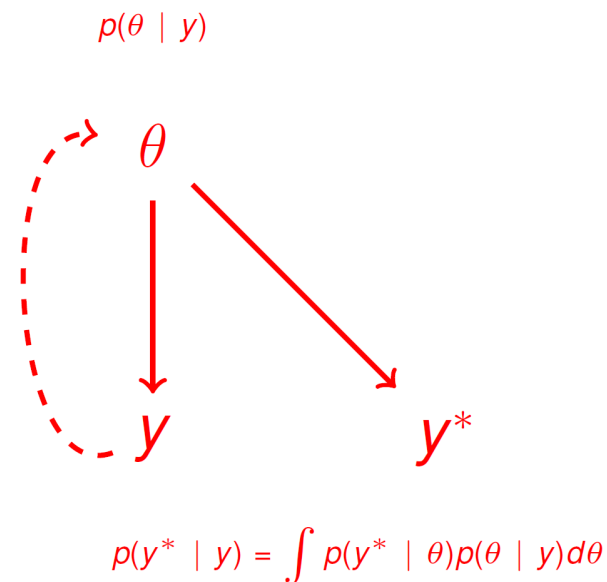
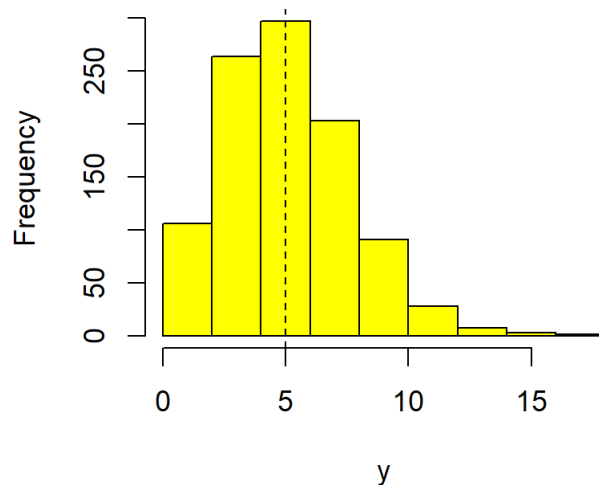
2 Prior: $\theta \sim \text{Beta}(1, 1)$

3 Posterior: $\theta \mid y \sim \text{Beta}(10 + 1, 58 - 10 + 1)$

4 PPD: $y^* \sim \text{Binomial}(\theta \mid y, 30)$

5 $P(y \geq 5) = \sum_{j=5}^{30} P(y^* = j)$

NOTE: we draw a new observation from a Binomial for each simulated model parameter



Introduction to Bayesian computing: Monte Carlo simulations

Bayesian computing

- In Session 2.1 we have introduced the concept of **conjugacy**, and we said that if the prior and posterior come from the same family of distributions, the prior is said to be **conjugate** to the likelihood → the posterior is a known distribution.
- In real life it is (almost) impossible to use conjugacy so we need to resort to simulative approaches or approximations to perform computation:
 - Monte Carlo methods
 - Markov Chain Monte Carlo (MCMC) methods
 - Integrated Nested Laplace Approximation (INLA)

Monte Carlo simulation

- A Monte Carlo (MC) simulation is a randomly evolving simulation.
- MC sampling is based on the idea that if you have a large random sample from a certain distribution, the statistics that you can calculate in this sample (mean, standard deviation, percentiles...) will be very similar to the corresponding theoretical values in the distribution.
- If you have a complicated mathematical expression for a distribution and you cannot calculate algebraically important parameters, you could get the computer to generate a large random sample from such a distribution.
- By calculating the mean of that parameter in the sample you could estimate the mean in the original distribution with great precision.

Example: a Monte Carlo approach to estimating tail-areas of distributions

Suppose we want to know the probability of getting 8 or more heads when we toss a fair coin 10 times.

- An *algebraic* approach would be:

y = Number of heads

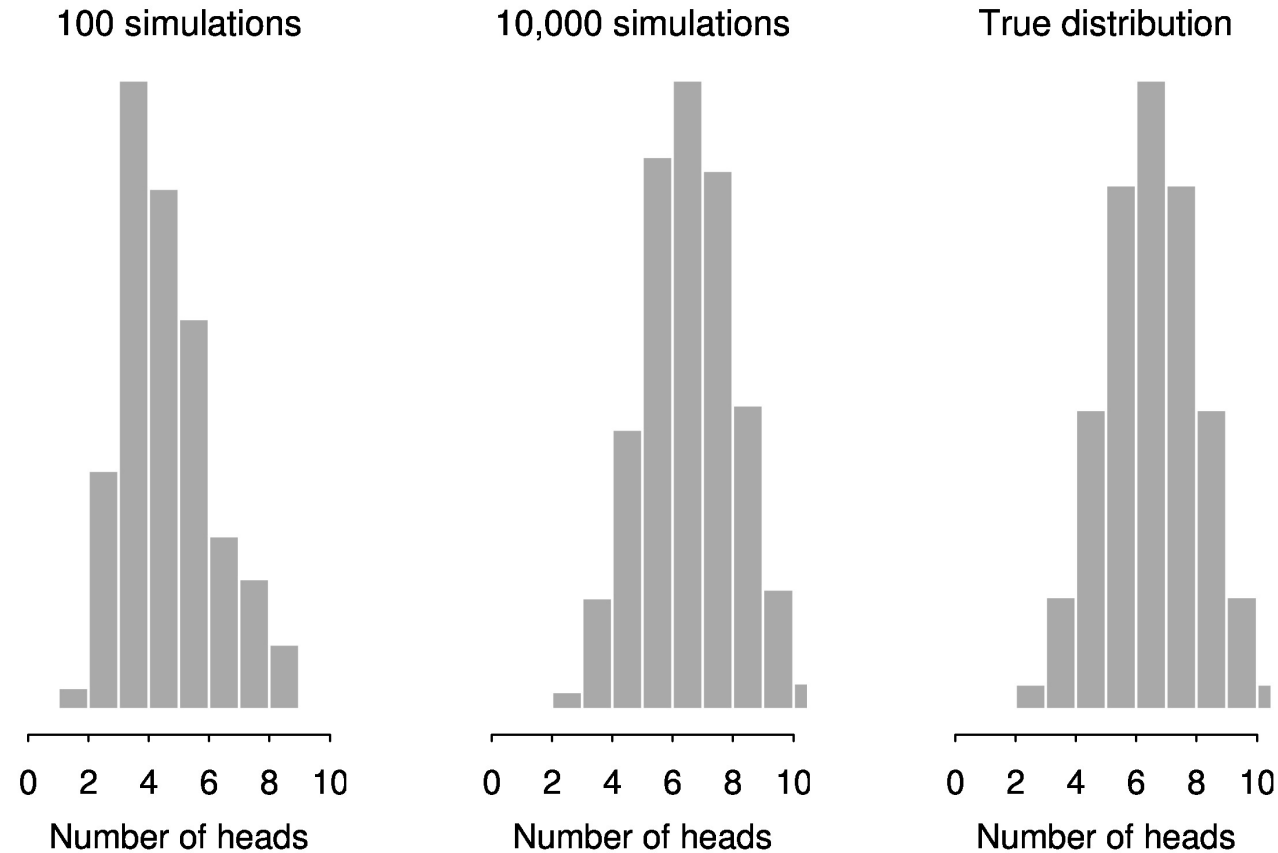
$y \sim \text{Binomial}(\theta, n)$

$$\begin{aligned} \Pr(\geq 8 \text{ heads}) &= \sum_{y=8}^{10} p\left(y \mid \theta = \frac{1}{2}, n = 10\right) \\ &= \binom{10}{8} \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^2 + \binom{10}{9} \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^1 + \binom{10}{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^0 \\ &= 0.0547 \end{aligned}$$

- A *physical* approach would be to repeatedly throw a set of 10 coins and count the proportion of throws that there were 8 or more heads.

Remember Binomial distribution: $p(y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$

- A *simulation* approach uses a computer to toss the coins!



Proportion with 8 or more heads in 10 tosses: (a) After 100 throws (0.02); (b) after 10,000 throws (0.0577); (c) the true Binomial distribution (0.0547).

Monte Carlo approach to approximate log-odds

- We start with a Binomial likelihood

$$y \mid \theta \sim \text{Binomial}(\theta, n)$$

combined with a

$$\text{Beta}(a, b)$$

as prior for the probability of success θ .

- We are interested in the log-odds function of θ defined as

$$\log \left(\frac{\theta}{1 - \theta} \right)$$

- The integral

$$\int_0^1 \log \left(\frac{\theta}{1 - \theta} \right) p(\theta \mid y) d\theta$$

cannot be computed analytically; we resort to Monte Carlo approximation.

Example of MC in practice

- We simulate m independent values $\{\theta^{(1)}, \dots, \theta^{(m)}\}$ from the

$$\text{Beta}(a_1 = y + a, b_1 = n - y + b)$$

posterior distribution using the property of conjugacy (Beta prior is conjugate to the Binomial likelihood).

- We apply the log-odds transformation to each value obtaining the set of values

$$\left\{ \log \left(\frac{\theta^{(1)}}{1 - \theta^{(1)}} \right), \dots, \log \left(\frac{\theta^{(m)}}{1 - \theta^{(m)}} \right) \right\}$$

- Finally, we compute the sample mean

$$\frac{\sum_{i=1}^m \log \left(\frac{\theta^{(i)}}{1 - \theta^{(i)}} \right)}{m}$$

which is the Monte Carlo approximation to

$$\log \left(\frac{\theta}{1 - \theta} \right)$$

Example of MC: R code

In R:

```
> a <- 1      Positive shape parameters
> b <- 1
> theta <- rbeta(1,a,b)
> n <- 1000
> y <- rbinom(1, size=n, p=theta)
```

number of observations

number of trials

probability of success on each trial

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> a <- 1
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```

- With this setting the exact posterior distribution of θ is

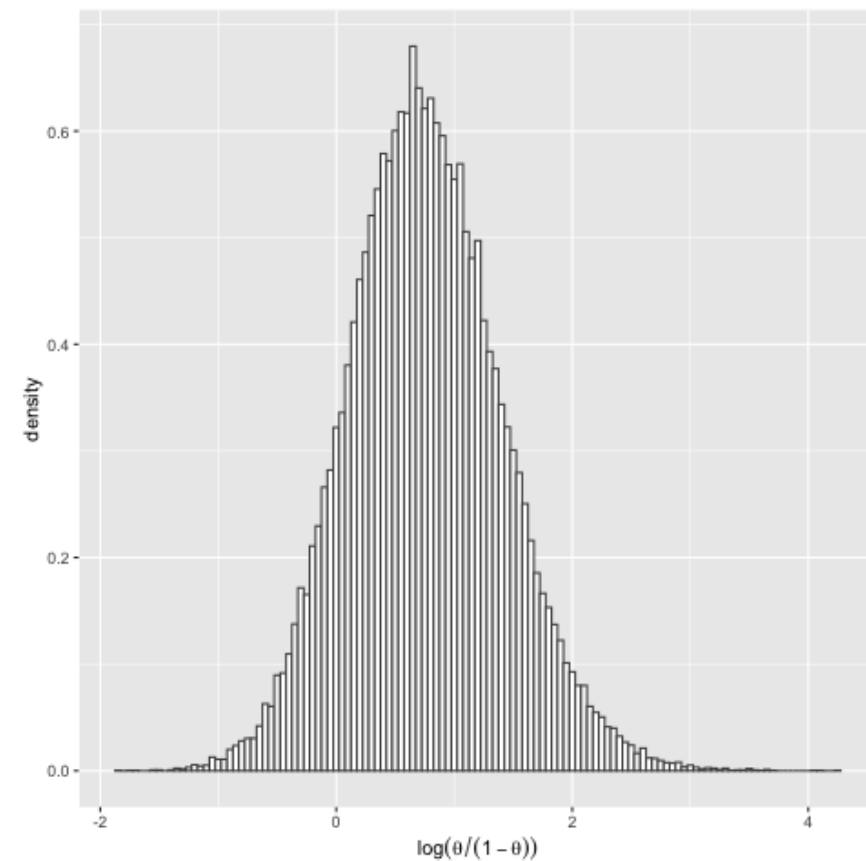
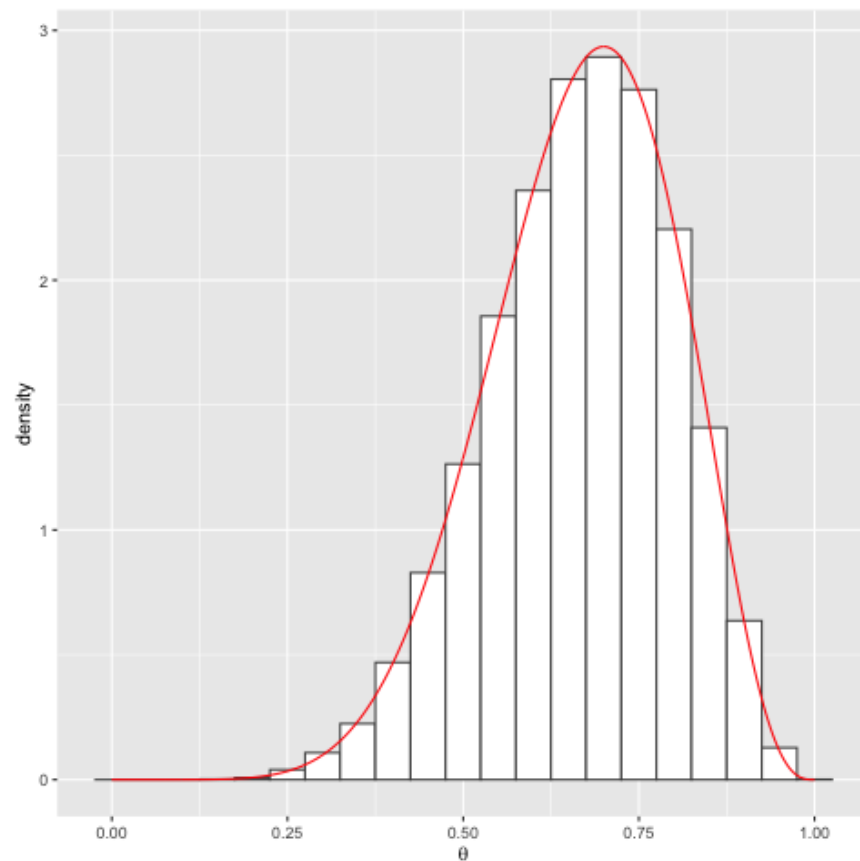
$$\text{Beta}(a_1 = a + y, b_1 = n - y + b)$$

- To approximate the log-odds, we simulate $m = 50000$ values from this Beta posterior distribution using the `rbeta` function.

```
> a1 <- a + y
> b1 <- n - y + b
> sim <- rbeta(n=50000, shape1=a1, shape2=b1)
> logodds <- log(sim/(1-sim))
```


Results and comparison with the theoretical distribution

The empirical distribution of the Monte Carlo sample is plotted below together with the exact posterior distribution of θ .



This is the histogram of the empirical distribution of the MC sample of log odds with $m = 50000$